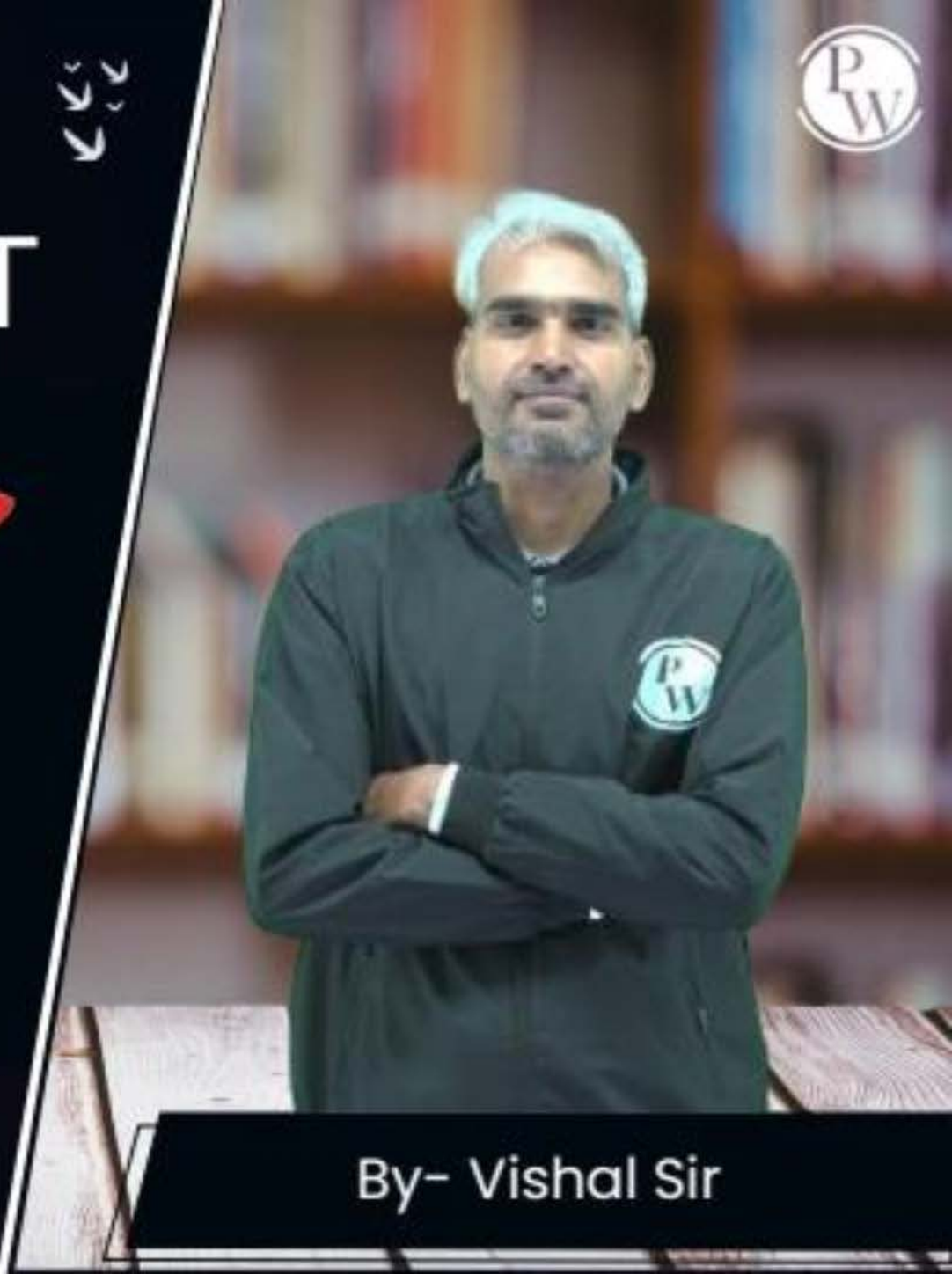
Computer Science & IT



Database Management System

Query Languages

Lecture No. 08



By- Vishal Sir

Recap of Previous Lecture



- **Topic** SQL clauses
- **Topic** Introduction to nested query
- **Topic** IN, ANY, ALL and EXISTS operators

Slide

Topics to be Covered



- Topic: AS clause
- Topic: WITH clause
- Topic: Comparison with NULL

Slide



Topic : Order of execution

Order of Execution:-

1. From
2. Where
3. Group By
4. Having
5. Select
6. Order BY



Topic : Nested queries

"Sub-query"

```
Select  Sid
From  Student
Where ( Marks = (Select  Max (Marks)
                 From  Student) )
```

Outer query (Main query)

it will select the maximum marks from Complete Student table

inner query (Sub-query)

Nested Query

Independent Nested query (Independent Sub-query)

* If inner query can be executed independently, then it is called independent Sub-query

Correlated Nested query (Correlated Sub-query)

* When execution of inner query requires value of an attribute from the relation specified in Outer query. then it is called Correlated Sub-query

Nested Query

Independent Nested query (Independent Sub-query)

eg:
```
Select Sid
From Student
Where (Marks) = (Select Max(Marks)
                 From Student)
```

it can be executed independently
∴ Independent Sub-query

Correlated Nested query (Correlated Sub-query)

eg:
```
Select R.A
From R
Where ....( Select *
            From S
            Where (R.B = S.C))
```

'R' is the relation specified in outer query

Outer query

It can not be executed independently, because it requires an attribute from the relation specified in the outer query

⇒ Hence, Correlated Sub-query

(1) order of execution w.r.t. independent Sub-query

Inner query will be executed first
And it will produce its output,
then outer query will use the output
produced by inner query for its
execution.

✱ Retrieve Sids of all the students Who scored maximum Marks.

inner query will execute first, and it will Produce 60 as o/p

* Select Sid
  from Student
  Where (Marks = 60 (Select Max(Marks) From Student))

↳ o/p =

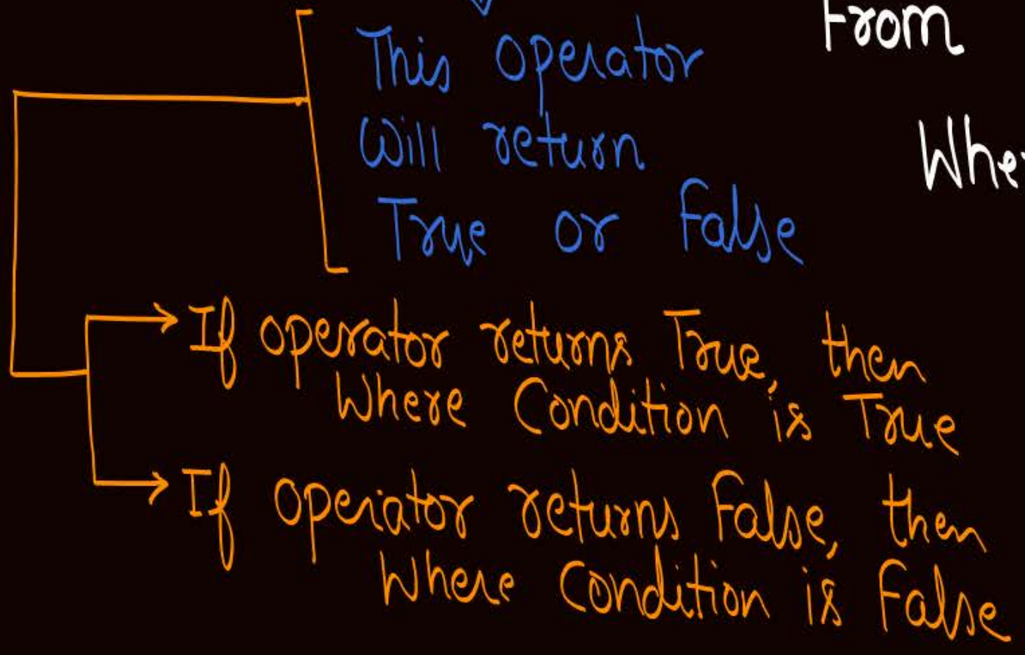
Sid
S3
S4

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

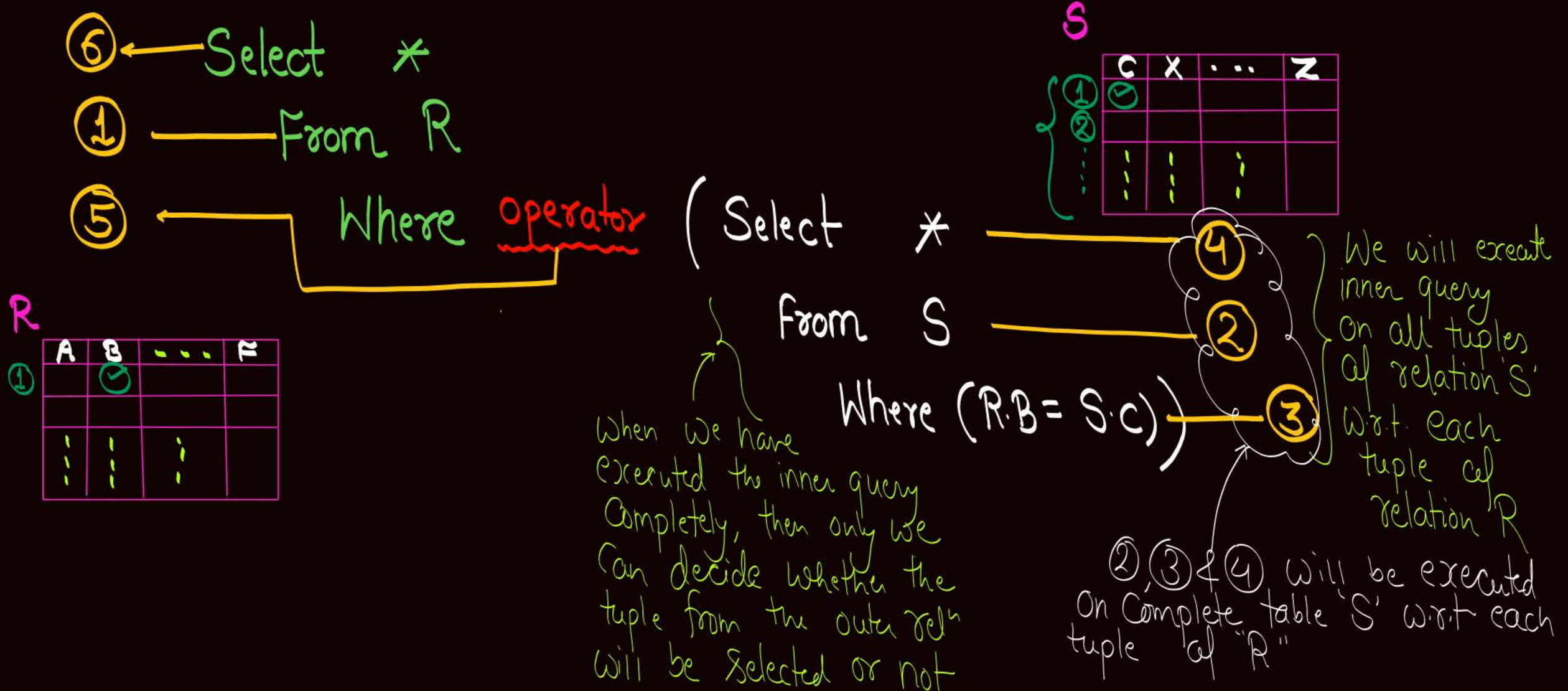
Correlated Sub-query :-

```
Select *
From R
Where operator ( Select *
                 From S
                 Where (R.B = S.C))
```



∴ Correlated Sub-query

Order of Execution W.r.t. Correlated Sub-query :-





Topic : Operators

* IN, ANY, ALL and EXISTS are the operators that are generally used with Sub-query Concept

IN operator

Operator "IN" is used to check whether the concerned tuple is a member of a set of tuples produced by the inner query or not?

Complement of 'IN' is 'NOT IN'

Check whether

X IN { 2, 5, 8, 9, 12, 13 }

Set of tuples (i.e, o/p produced by inner query)

Select Eid From Emp

it is our Concerned tuple from outer query

(i) If X = 8, then 'IN' will return True

(ii) If X = 10, then 'IN' will return False

Check whether

X IN {(1, A), (3, D), (4, D), (7, F)}

Set of tuples (i.e. O/p produced by inner query)

Select Eid, Ename From EMP

it is our Concerned tuple from outer query

(i) If X = (4, D) then 'IN' will return True

(ii) If X = (F, 7) then 'IN' will return False

(F, 7) ≠ (7, F)

Note :-

If set of tuples is Empty {i.e, o/p produced by inner query is Empty}, then operator 'IN' will always return False

and hence operator 'NOT IN' will always return true if inner query result is Empty.

Q: Consider the following relational scheme

Supplier (Sid, Sname, Rating)

Parts (Pid, Pname, Color)

Catalog (Sid, Pid, Cost)

* Retrieve Sids of suppliers who supplied some Red Color Parts

R.A:-

$$\pi_{Catalog.Sid}\left(\sigma_{Catalog.Pid = Parts.Pid \wedge Parts.Color = 'Red'}(Catalog \times Parts)\right)$$

SQL: (without IN)

```
Select C.Sid
    From Catalog C, Parts P
        Where ( C.Pid = P.Pid
                AND
            P.Color = 'Red' )
```

SQL: (Using IN)

```
Select Sid From Catalog
  Where Pid IN (Select Pid From Parts
                  Where Color = 'Red')
```

'ANY' operator

Operator 'ANY' is used along with Comparison operators

$<, \leq, >, \geq, =, <>$

Check whether w.r.t. 20< × × × × × × ← No one satisfies Comparison Condn

X < ANY { 2, 5, 8, 9, 12, 14 }

w.r.t. '10<' × × × × ✓

└ we don't need to Compare with remaining elements

if X = 10, then 'ANY' will return True

if X = 20, then 'ANY' will return false

'ANY' operator

Operator 'ANY' is used along with Comparison operators

$<, \leq, >, \geq, =, <>$

Note: Operator 'ANY' will return True, if and only if at least one tuple in the set of tuples satisfy the Comparison Condition

Note: If set of tuples is empty, then operator 'ANY' will always return False

'ALL' operator

Operator 'ALL' is also used along with Comparison operators

$<, \leq, >, \geq, =, <>$

Check whether W.r.t. '1<' ✓ ✓ ✓ ✓ ✓ ✓ { No one failed the Comparison Condn.

X < ALL { 2, 5, 8, 9, 12, 14 }

W.r.t. '10<' X → we don't need to Compare with remaining elements

if X = 10, then 'ALL' will return False

if X = 1, then 'ALL' will return True

'ALL' operator

Operator 'ALL' is also used along with Comparison operators

$<, \leq, >, \geq, =, <>$

- operator 'ALL' can return false if and only if at least one tuple in the set of tuples fails the Comparison Condition
- If set of tuples is Empty, then operator 'ALL' will always return True

EXISTS operator

- EXISTS returns True if and only if inner query result is not Empty.
  - If inner query result is not Empty, then EXISTS will return True

&

If inner query result is Empty, then EXISTS will return False.

Complement of 'EXISTS' is 'NOT EXISTS'

H.W. What output will be produced by the following query if query is executed on the following instances of Catalog and Parts table

#e.g.

```
SELECT C.sid
FROM Catalog C
WHERE EXISTS ( SELECT *
               FROM Parts P
               WHERE (P.Pid=C.Pid AND P.color='RED'))
```

i. Correlated Sub-query

Parts

Pid	Color
P1	Red
P2	Green
P3	Red

Catalog

Sid	Pid
S1	P1
S1	P2
S2	P1
S2	P2
S3	P1
S3	P3

Select C.Sid
From Catalog C
Where EXISTS (Select P1.* Red
From Parts P
Where (P.Pid = C.Pid
AND
P.color = 'Red'))

Catalog

	Sid	Pid
①	S_1	P_1
②	S_1	P_2
③	S_2	P_2
④	S_3	P_2
⑤	S_3	P_3

O/P =
S_1 ✓

	C.Sid	C.Pid
①	S_1	P_1

Parts

Pid	Color
P_1 ✓	Red ✓
P_2 ✗	Green
P_3 ✗	Red

Not Empty

∴ EXISTS will return True

i.e, Where Cond^{n} is true.

Hence Sid of the first tuple of Catalog is selected

Select C.Sid
From Catalog C
Where EXISTS (Select *
From Parts P
Where (P.Pid = C.Pid
AND
P.color = 'Red'))

Catalog

	Sid	Pid
①	S_1	P_1
②	S_1	P_2
③	S_2	P_2
④	S_3	P_2
⑤	S_3	P_3

O/P =
S_1 ✓
~~S_1~~

② C.Sid C.Pid: S_1 P_2

Parts

Pid	Color
P_1 ×	Red
P_2 ✓	Green ×
P_3 ×	Red

Empty

∴ EXISTS will return False

i.e, Where Cond^{n} is false

Hence Sid of the 2nd tuple of Catalog is not selected

Select C.Sid
From Catalog C
Where EXISTS (Select *
From Parts P
Where (P.Pid = C.Pid
AND
P.color = 'Red'))

③ C.Sid C.Pid
S_2 P_2

Parts

Pid	Color
P_1	Red
P_2	Green
P_3	Red

Catalog

	Sid	Pid
①	S_1	P_1
②	S_1	P_2
③	S_2	P_2
④	S_3	P_2
⑤	S_3	P_3

O/P = S_1 ~~S_1~~ ~~S_2~~

Empty

∴ EXISTS will return False

i.e, Where Cond^n is false

Hence Sid of the 3rd tuple of Catalog is not selected

Select C.Sid
From Catalog C
Where EXISTS (Select *
From Parts P
Where (P.Pid = C.Pid
AND
P.color = 'Red'))

Empty

∴ EXISTS will return false

i.e, Where Cond^{n} is false

Hence Sid of the 4th tuple of Catalog is not selected

	C.Sid	C.Pid
④	S_3	P_2

Parts

Pid	Color
P_1	Red
P_2	Green
P_3	Red

Catalog

	Sid	Pid
①	S_1	P_1
②	S_1	P_2
③	S_2	P_2
④	S_3	P_2
⑤	S_3	P_3

O/P = S_1 ~~S_1~~ ~~S_2~~ ~~S_3~~

Select C.Sid
From Catalog C
Where EXISTS (Select P3 * Red
From Parts P
Where (P.Pid = C.Pid
AND
P.color = 'Red'))

C.Sid C.Pid

⑤ | S_3 | P_3 |

Parts

Pid	Color
P_1	Red
P_2	Green
P_3	Red

Catalog

	Sid	Pid
①	S_1	P_1
②	S_1	P_2
③	S_2	P_2
④	S_3	P_2
⑤	S_3	P_3

O/P = S_1 S_1 S_2 S_3 S_3

Not Empty

∴ EXISTS will return True

i.e, Where Cond^{n} is True

Hence Sid of the 5th tuple of Catalog is selected

```
Select C.Sid
From Catalog C
Where EXISTS (Select *
              From Parts P
              Where (P.Pid = C.Pid
                     AND
                     P.color = 'Red'))
```

Parts

Pid	Color
P_1	Red
P_2	Green
P_3	Red

Catalog

	Sid	Pid
①	S_1	P_1
②	S_1	P_2
③	S_2	P_2
④	S_3	P_2
⑤	S_3	P_3

o/p =

Sid
S_1
S_3

Above query will select the Sids of the Suppliers Who have supplied some Red Color Parts.

H.W.

#e.g.

```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS ( SELECT P.Pid
                   FROM Parts P
                   WHERE NOT EXISTS (SELECT C2.Sid
                                     FROM Catalog C2
                                     WHERE(C2.Pid=P.Pid AND  C2.Sid=C1.Sid)))
```

Catalog

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2

Parts

Pid
P_1
P_2

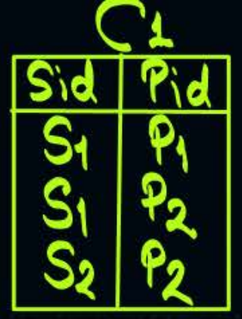
What output is produced by above SQL query:

A. Sids of suppliers who supplied some parts o/p = $\{S_1, S_2\}$

B. Sids of suppliers who supplied only proper subset of parts from all parts o/p = $\{S_2\}$

C. Sids of suppliers who supplied all parts o/p = $\{S_1\}$

D. Sids of suppliers who did not supply any part { Can never be true w.r.t. given query }

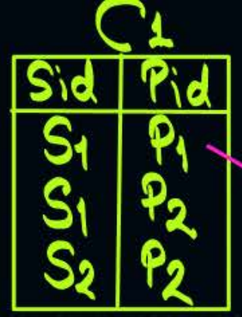




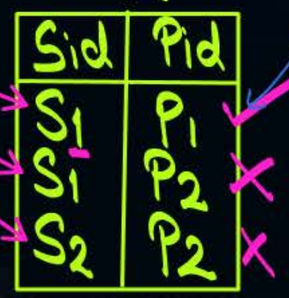


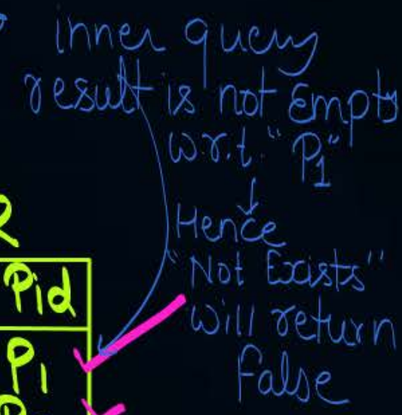

```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS (SELECT P.Pid
                  FROM Parts P
                  WHERE NOT EXISTS (SELECT C2.Sid
                                    FROM Catalog C2
                                    WHERE(C2.Pid=P.Pid AND  C2.Sid=C1.Sid)))
```

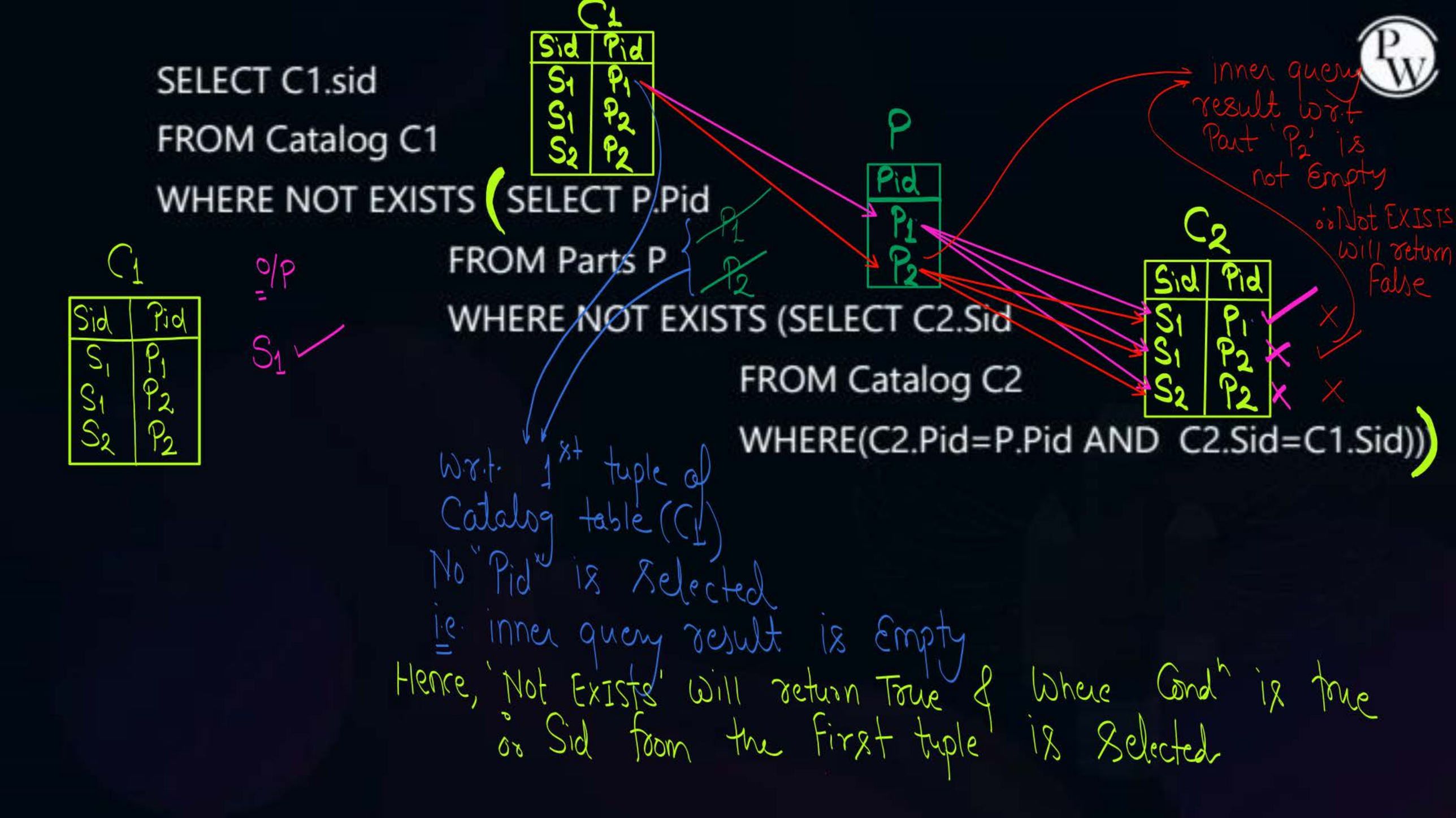
```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS (SELECT P.Pid
                  FROM Parts P
                  WHERE NOT EXISTS (SELECT C2.Sid
                                    FROM Catalog C2
                                    WHERE(C2.Pid=P.Pid AND  C2.Sid=C1.Sid)))
```

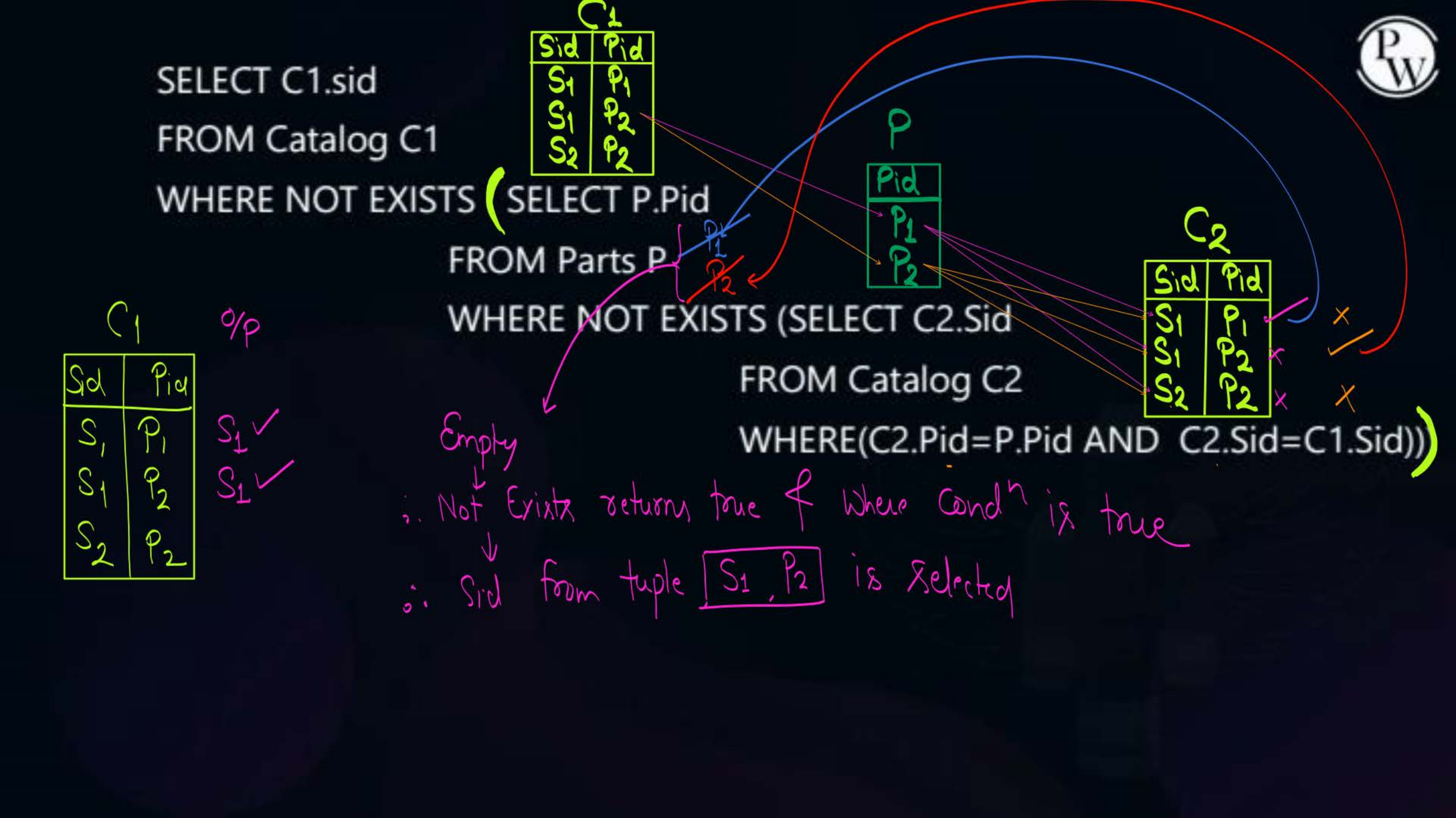


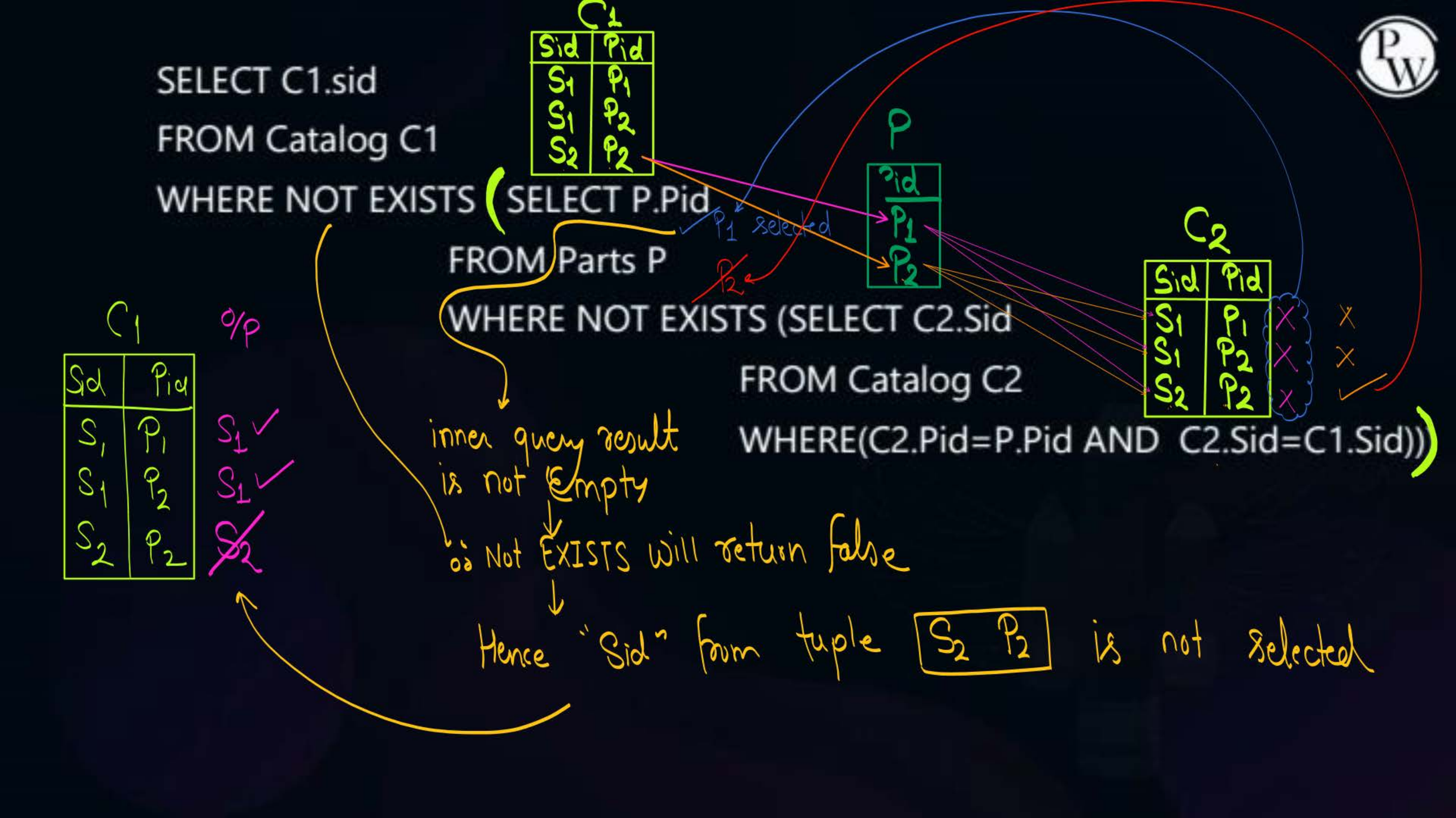







C1
Sid Pid
S1 P1
S1 P2
S2 P2
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS (SELECT P.Pid
FROM Parts P
WHERE NOT EXISTS (SELECT C2.Sid
FROM Catalog C2
WHERE(C2.Pid=P.Pid AND C2.Sid=C1.Sid))
P
Pid
P1
P2
P1
P2
C2
Sid Pid
S1 P1
S1 P2
S2 P2
Inner query result w.r.t Part 'P2' is not Empty
∴Not EXISTS will return False
C1
Sid Pid
S1 P1
S1 P2
S2 P2
O/P
S1
W.r.t. 1st tuple of Catalog table (C1)
No "Pid" is Selected
i.e. inner query result is Empty
Hence, 'Not EXISTS' will return True & Where Cond^n is true
∴ Sid from the first tuple is Selected


SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS (SELECT P.Pid
FROM Parts P
WHERE NOT EXISTS (SELECT C2.Sid
FROM Catalog C2
WHERE(C2.Pid=P.Pid AND C2.Sid=C1.Sid))
C1
Sid Pid
S1 P1
S1 P2
S2 P2
P
Pid
P1
P2
C2
Sid Pid
S1 P1
S1 P2
S2 P2
C1
Sid Pid
S1 P1
S1 P2
S2 P2
O/P
S1
S1
Empty
∴ Not Exists returns true & Where Condn is true
∴ Sid from tuple S1 , P2 is selected


SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS (SELECT P.Pid
FROM Parts P
WHERE NOT EXISTS (SELECT C2.Sid
FROM Catalog C2
WHERE(C2.Pid=P.Pid AND C2.Sid=C1.Sid))
C1
Sid Pid
S1 P1
S1 P2
S2 P2
P
Pid
P1
P2
P1 selected
P2
C2
Sid Pid
S1 P1
S1 P2
S2 P2
C1
Sid Pid
S1 P1
S1 P2
S2 P2
O/P
S1
S1
inner query result is not Empty
∴ Not EXISTS will return false
Hence "Sid" from tuple S2 P2 is not selected

H.W.

#e.g.

```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS ( SELECT P.Pid
                   FROM Parts P
                   WHERE NOT EXISTS (SELECT C2.Sid
                                     FROM Catalog C2
                                     WHERE(C2.Pid=P.Pid AND  C2.Sid=C1.Sid)))
```

Catalog

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2

Parts

Pid
P_1
P_2

What output is produced by above SQL query:

A. Sids of suppliers who supplied some parts o/p = $\{S_1, S_2\}$

B. Sids of suppliers who supplied only proper subset of parts from all parts o/p = $\{S_2\}$

C. Sids of suppliers who supplied all parts o/p = $\{S_1\}$

D. Sids of suppliers who did not supply any part {Can never be true w.r.t. given query}



Topic : AS clause

AS clause is used to rename a column or table with an alias. {Note: AS Clause can be used to give a new_name (alias) to almost Everything}

- An alias only exists for the duration of the query.

#e.g. Consider a database that has the relation schema EMP (EmpId, EmpName, [redacted] DeptName).
An instance of the schema EMP and a SQL query on it are given below:

EmpId	Emp Name	DeptName
1	XYA	AA
2	XYB	AA
3	XYC	AA
4	XYD	AA
5	XYE	AB
6	XYF	AB
7	XYG	AB
8	XYH	AC
9	XYI	AC
10	XYJ	AC
11	XYK	AD
12	XYL	AD
13	XYM	AE

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
    (SELECT DeptName, COUNT(EmpId) AS EC(DeptName, Num)
                                    FROM EMP
                                    GROUP BY DeptName)
```

The output of executing the SQL query is_____

#e.g. Consider a database that has the relation schema EMP (EmpId, EmpName, and DeptName).
An instance of the schema EMP and a SQL query on it are given below:

Outer query

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
    (SELECT DeptName, COUNT(EmpId) AS EC(DeptName, Num)
                                    FROM EMP
                                    GROUP BY DeptName)
```

Select DeptName, Count(EmpId) AS EC(DeptName, Num)

EC

DeptName	Num
DeptName	Count(EmpID)

o/p =

The output of executing the SQL query is _____

inner query

#e.g. Consider a database that has the relation schema
EMP (EmpId, EmpName, and DeptName).
An instance of the schema EMP and a SQL query on it are given below:

Outer query

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
    (SELECT DeptName, COUNT(EmpId) AS EC(DeptName, Num)
                                    FROM EMP
                                    GROUP BY DeptName)
```

The output of executing the SQL query is _____

inner query

↓

it is independent inner query

∴ inner query will be executed first,

Because of execution of inner query relation EC (DeptName, Num) will come into the picture

It will exist only during the Execution of overall query

#e.g. Consider a database that has the relation schema
EMP (EmpId, EmpName, and DeptName).
An instance of the schema EMP and a SQL query on it are given below:

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
    (SELECT DeptName, COUNT(EmpId) AS EC(DeptName, Num)
                                    FROM EMP
                                    GROUP BY DeptName)
```

The output of executing the SQL query is_____



2 mins Summary

Topic AS clause

Topic WITH clause

Topic Comparison with NULL

Slide

THANK - YOU

Slide